

AgriNet AI

Interoperable Digital Agriculture Intelligence Platform

Architecture & Solution Document

1. Executive Summary

AgriNet AI is an interoperable digital agriculture intelligence platform inspired by the BRICS AgriN initiative.

The platform is designed to empower small and marginal farmers with **localized, data-driven and AI-powered agricultural intelligence** by combining farmer profiles, soil health information, weather intelligence, satellite/NDVI indicators and AI-based crop disease detection.

Google Gemini / Vertex AI acts as the core reasoning layer, transforming multi-source agricultural data into actionable recommendations.

The architecture is designed as a scalable **Digital Public Good (DPG)** that can be adapted across countries, languages, crops, agricultural practices and climatic conditions.

2. Problem Statement

Small and marginal farmers frequently lack access to timely and localized agricultural intelligence.

Traditional agricultural decisions may depend heavily on:

- Generalized farming practices
- Limited soil information
- Unpredictable weather
- Delayed identification of crop-health problems
- Fragmented government scheme information
- Limited access to agricultural experts
- Language and technology barriers

This creates challenges related to productivity, resource utilization, climate resilience and long-term soil health.

AgriNet AI addresses this challenge by connecting multiple intelligence sources through a single AI-powered advisory platform.

3. Proposed Solution

AgriNet AI follows a multi-source intelligence architecture.

The farmer interacts through a web or voice interface.

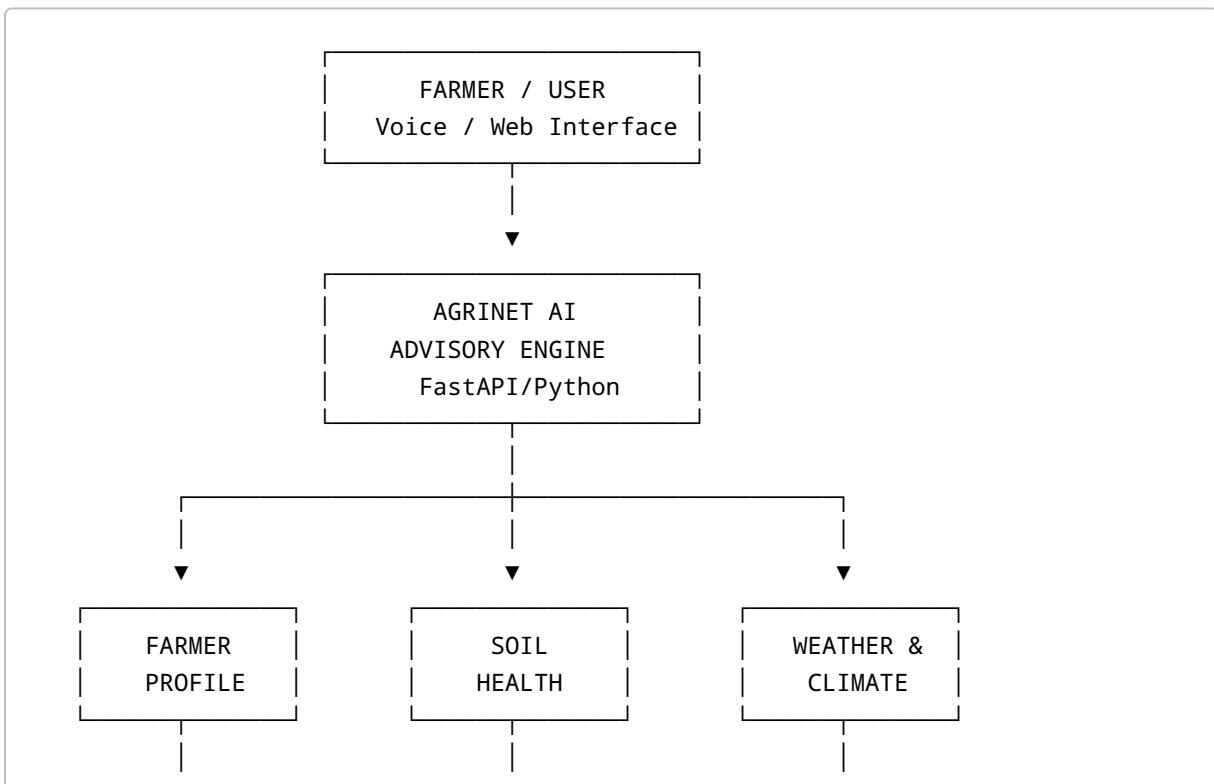
The platform combines:

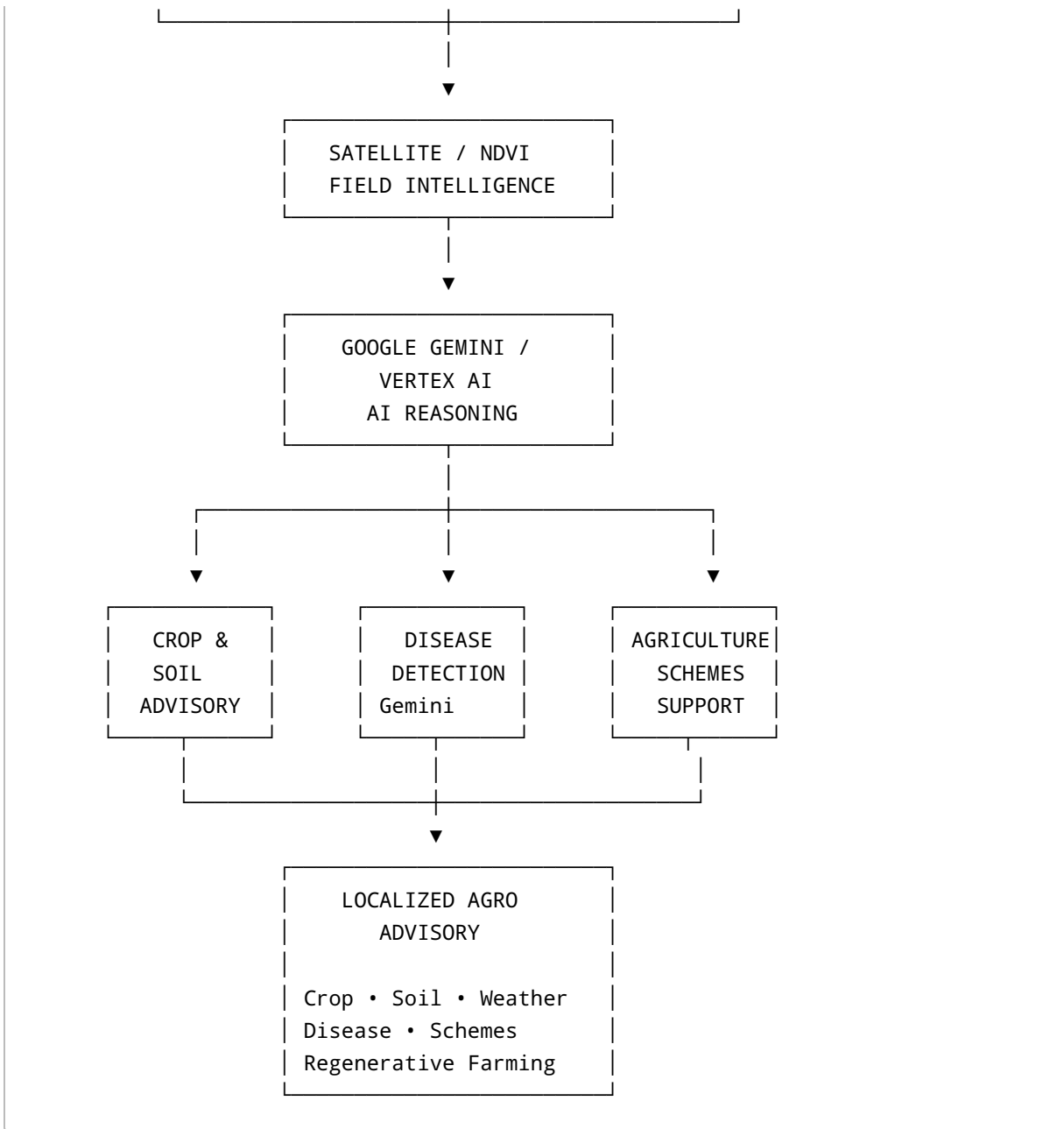
- Farmer profile and farm context
- Crop information
- Soil health indicators
- Weather and climate information
- Satellite/NDVI intelligence
- Crop/leaf images
- Agricultural scheme information

These inputs are processed through the AgriNet AI advisory engine and provided to **Google Gemini / Vertex AI** for contextual reasoning.

The resulting output is converted into a localized agricultural advisory.

4. High-Level Architecture





5. Technology Architecture

Frontend

The AgriNet AI frontend provides the farmer-facing interaction layer.

Key capabilities

- Web-based farmer interface
- Voice-enabled interaction
- Farmer profile visualization
- Soil intelligence display
- Weather intelligence
- Satellite/NDVI indicators
- Crop disease image upload
- AI advisory generation
- Agricultural scheme information

The frontend communicates with the backend advisory APIs.

6. Backend

The backend is implemented using **Python and FastAPI**.

Primary responsibilities

- API request handling
- Farmer profile retrieval
- Agricultural data aggregation
- Soil intelligence processing
- Weather data integration
- Satellite intelligence integration
- AI advisory orchestration
- Crop disease analysis
- Agricultural scheme retrieval

Core backend technologies

- Python
 - FastAPI
 - Uvicorn
 - Pydantic
 - Requests
 - python-dotenv
 - Google GenAI SDK
-

7. AI Reasoning Layer

Google Gemini / Vertex AI

Google Gemini / Vertex AI provides the reasoning capability of the platform.

Instead of relying on a single data source, the AI receives contextual agricultural information.

Example

```
Farmer Profile
  +
Crop
  +
Soil Health
  +
Weather
  +
Satellite / NDVI
  +
Farm Conditions
  ↓
Gemini / Vertex AI
  ↓
Localized Agricultural Advisory
```

This allows the system to move from **generic information retrieval to contextual agricultural intelligence**.

8. Farmer Profile Intelligence

AgriNet AI uses structured farmer profiles.

Example — India

Farmer: Ravi Kumar
Farmer ID: farmer001
Country: India
State: Rajasthan
District: Jaipur
Language: Hindi
Crop: Wheat
Land: 5 acres

Irrigation: Canal

Season: Rabi

Soil Profile

Parameter	Value
pH	7.8
Nitrogen	Low
Phosphorus	Medium
Potassium	Medium
Organic Carbon	Low
Moisture	32.5%

This information provides the AI with contextual information before generating recommendations.

9. BRICS Country Data Model

The platform demonstrates how the same architecture can support multiple countries.

India

- Region: Jaipur
- Language: Hindi
- Currency: INR
- Example crops: Wheat, Millet, Mustard, Pulses

Brazil

- Region: Mato Grosso
- Language: Portuguese
- Currency: BRL
- Example crops: Soybean, Maize, Beans, Sorghum

Russia

- Region: Krasnodar
- Language: Russian
- Currency: RUB
- Example crops: Wheat, Barley, Sunflower, Rapeseed

China

- Region: Heilongjiang
- Language: Chinese
- Currency: CNY
- Example crops: Rice, Maize, Soybean, Wheat

South Africa

- Region: Free State
- Language: English
- Currency: ZAR
- Example crops: Maize, Sorghum, Sunflower, Wheat

The country-specific information is separated from the core advisory architecture, allowing the platform to scale without redesigning the complete system.

10. Satellite Intelligence

Satellite intelligence provides spatial context for agricultural decision-making.

The prototype demonstrates satellite intelligence using **representative Sentinel-2 / NDVI-style data**.

Example indicators

- NDVI
- Crop health
- Vegetation condition
- Field condition
- NDVI trend

Prototype Disclaimer

The current satellite layer is a **representative prototype/demo layer** used to demonstrate the interoperable architecture.

It should not be represented as live Sentinel-2 or Google Earth Engine processing.

Future Enhancement

The satellite layer can be replaced or enhanced with **Google Earth Engine** and production geospatial processing pipelines.

11. Soil Health Intelligence

Soil health is a key component of regenerative agriculture.

The platform can incorporate:

- Soil pH
- Nitrogen
- Phosphorus
- Potassium
- Organic carbon
- Soil moisture

These parameters can influence:

- Crop recommendations
- Irrigation guidance
- Soil improvement practices
- Crop rotation
- Sustainable farming recommendations

Example

For the demo farmer Ravi Kumar:

- pH: 7.8
- Nitrogen: Low
- Organic carbon: Low
- Moisture: approximately 32%

The AI can use these indicators together with weather and crop context when generating an advisory.

12. Weather & Climate Intelligence

Weather information is incorporated into agricultural decision-making.

The platform can use:

- Current temperature
- Rainfall
- Rain probability
- Forecast conditions
- Climate-risk indicators

Weather intelligence can support decisions involving:

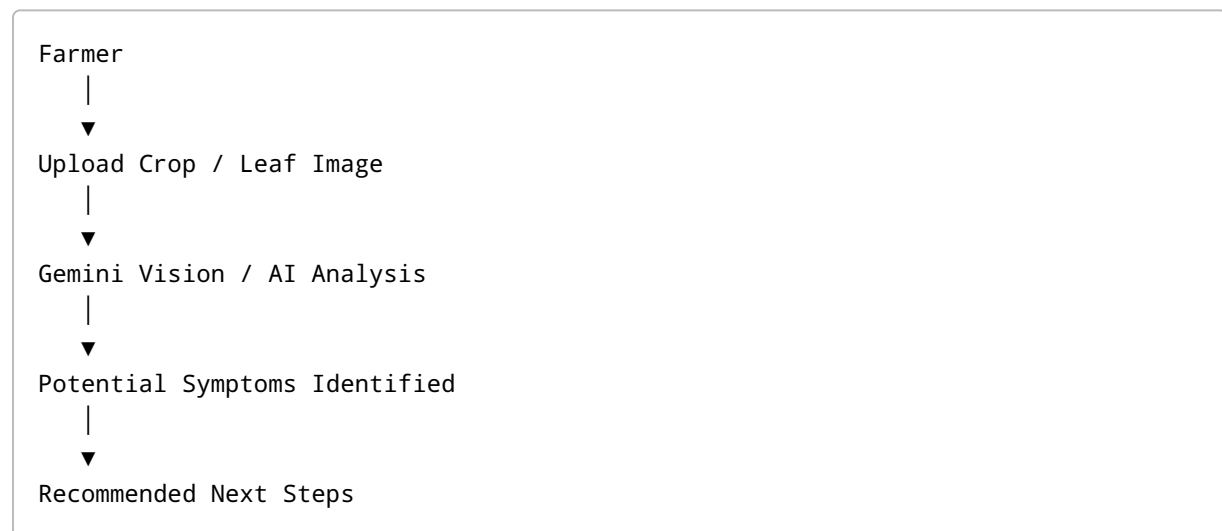
- Irrigation
- Rain-sensitive farming activities
- Crop protection
- Climate resilience
- Water conservation

The objective is to make recommendations that consider both **current farm conditions and environmental conditions**.

13. Crop Disease Detection

AgriNet AI includes AI-based crop-health screening.

Workflow



The system is designed to identify **potential crop-health symptoms**, rather than presenting an uncertain AI result as a guaranteed diagnosis.

This capability provides farmers with an additional early-warning mechanism.

14. Regenerative Agriculture

AgriNet AI is designed not only for short-term yield optimization but also for **long-term soil and ecosystem health**.

Recommendations may consider:

Soil Health

- Organic matter improvement
- Soil-friendly practices
- Balanced nutrient management

Crop Rotation

- Diversification
- Rotation planning
- Reduced soil depletion

Water Conservation

- Irrigation efficiency
- Moisture-aware recommendations
- Reduced unnecessary water usage

Climate Resilience

- Climate-aware crop selection
- Weather-risk management
- Resource-efficient practices

Sustainable Crop Selection

The AI can consider soil, climate and regional context when recommending suitable crops.

15. Agricultural Scheme Intelligence

The prototype includes a cross-country agricultural scheme data model.

India

PM-KISAN

Category: Farmer Income Support

Pradhan Mantri Fasal Bima Yojana

Category: Crop Insurance

Brazil

PRONAF

Category: Family Farming

South Africa

Farmer Support Programme

Category: Agricultural Support

Russia

Agricultural Producer Support

Category: Agricultural Support

China

Agricultural Producer Support

Category: Agricultural Support

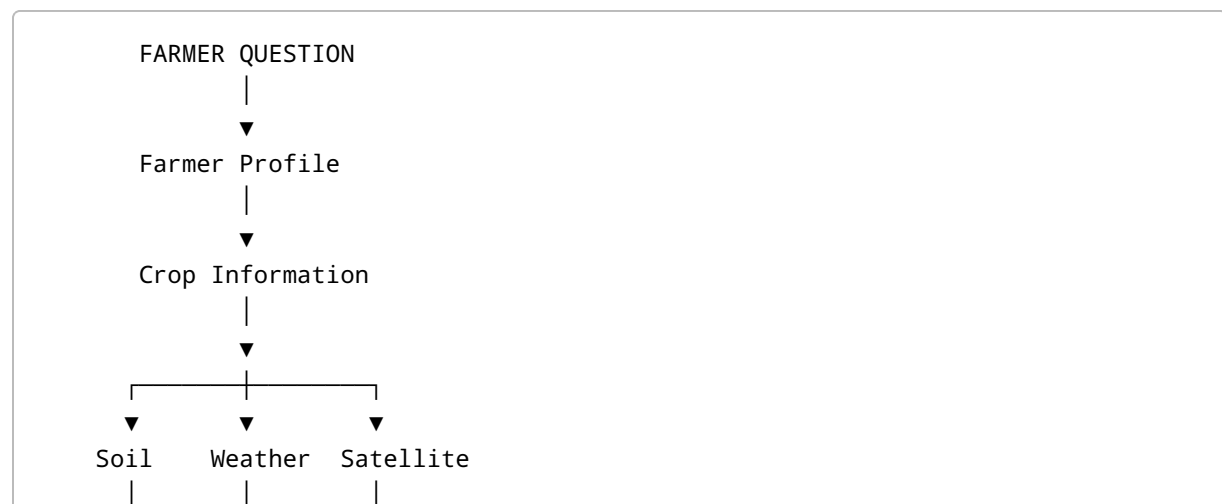
Important Data Qualification

The supplied scheme dataset is marked "**Reference Information.**"

For a production system, scheme eligibility, benefits, application procedures and current status should be connected to and verified against official government sources.

The AI should not promise scheme approval or eligibility.

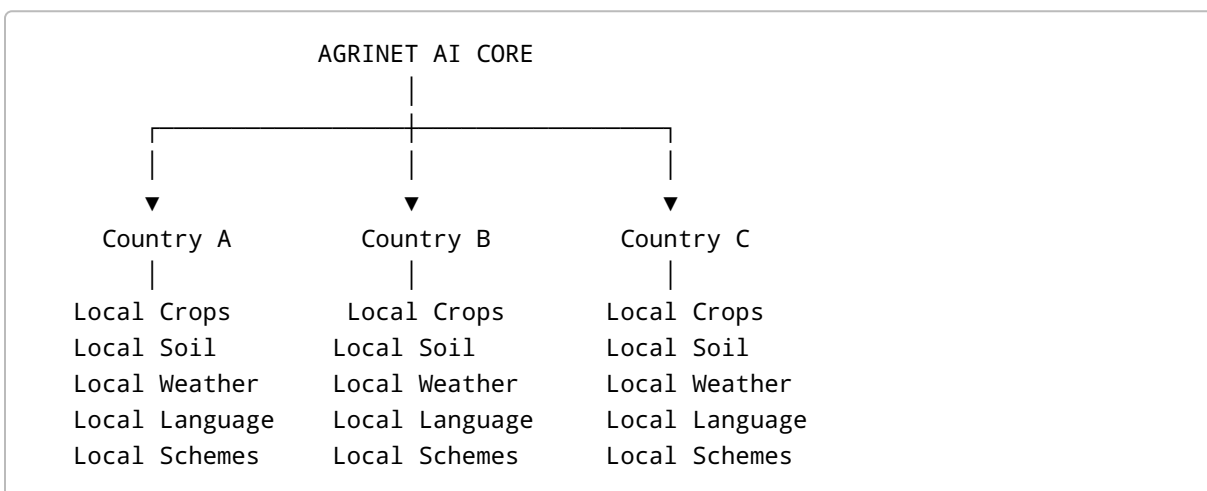
16. Advisory Generation Flow





17. Interoperability Model

AgriNet AI separates the **core intelligence architecture** from country-specific data.



This architecture allows the same platform to operate across different agricultural ecosystems.

18. Digital Public Good Vision

AgriNet AI is designed with Digital Public Good principles in mind.

Reusable

Agricultural advisory components can be adapted for different regions.

Interoperable

Different data providers can be connected through modular interfaces.

Multilingual

The farmer interaction layer can support different languages.

Scalable

The architecture can expand from prototype to production infrastructure.

Cross-Border

The same intelligence framework can support agricultural ecosystems across BRICS countries.

19. Current Prototype vs Future Production

Component	Current Prototype	Future
AI	Gemini / Vertex AI	Production-grade AI orchestration
Backend	FastAPI / Python	Scalable cloud deployment
Satellite	Representative Sentinel-2/NDVI data	Google Earth Engine / production satellite pipeline
Soil	Structured demo data	Live/verified soil datasets
Weather	Weather intelligence	Multi-provider climate intelligence
Disease	Gemini Vision	Specialized agricultural vision models
Schemes	Reference dataset	Verified government integrations
Languages	Country-specific model	Expanded multilingual/voice support
Deployment	Prototype	Production-scale infrastructure

20. Security & Reliability Considerations

For production deployment, AgriNet AI should include:

- API authentication
- Secure environment variables
- Input validation
- Rate limiting
- Data encryption
- Audit logging
- AI response monitoring
- Source attribution
- Agricultural data validation
- Government scheme verification
- Human escalation for uncertain/high-impact cases

AI-generated agricultural recommendations should clearly distinguish between **verified information**, **model-generated guidance** and **reference data**.

21. Future Roadmap

Phase 1 — Prototype

AI advisory + farmer profiles + soil + weather + representative satellite intelligence.

Phase 2 — Geospatial Intelligence

Integrate Google Earth Engine and production satellite data.

Phase 3 — Advanced Agricultural AI

Introduce specialized models for:

- Crop health
- Disease detection
- Yield intelligence
- Climate risk
- Soil health

Phase 4 — BRICS Expansion

Add country-specific:

- Agricultural datasets
- Languages

- Schemes
- Crops
- Weather models
- Regional advisory rules

Phase 5 — Digital Agriculture Ecosystem

Develop reusable APIs, data standards, AI models and advisory frameworks that can be adopted by agricultural organizations and government ecosystems.

22. Expected Impact

AgriNet AI aims to provide:

Sustainable Agriculture

Encourages regenerative and resource-efficient farming.

Data-Driven Decisions

Combines multiple agricultural intelligence sources.

AI-Powered Advisory

Converts complex data into understandable recommendations.

Crop Health Intelligence

Provides early screening of potential crop-health problems.

Climate Resilience

Makes agricultural decisions more weather- and climate-aware.

Farmer Accessibility

Uses multilingual and voice-enabled interaction.

Cross-Border Interoperability

Creates a foundation for agricultural intelligence across BRICS nations.

23. Conclusion

AgriNet AI transforms fragmented agricultural information into contextual, actionable intelligence.

By combining:

Farmer Context + Soil + Weather + Satellite + AI Vision + Gemini/Vertex AI

the platform creates a unified intelligence layer for modern agriculture.

Its modular architecture enables the solution to evolve from a prototype into a scalable **Digital Public Good for interoperable digital agriculture.**

AgriNet AI

From Data → Intelligence → Action → Sustainable Agriculture